

MD RAHBIR MAHDI

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EDUCATION

York University

Bachelor of Science in Computer Science

Toronto, ON

Sep. 2023 – Sep. 2027

EXPERIENCE

Co-Founder & Software Developer

Feb. 2026 – Present

☑ *Clarus (HackCanada 2026 Winner)*

NSU Startup Cohort

- Selected for the NSU **startup cohort 2026** to build **Clarus** into a production healthcare-automation platform, powered by **FastAPI**, **Next.js**, **React Flow**, **ElevenLabs**, **Twilio**, and **Supabase**.
- Won **Vivirion Solutions–Best Practical Healthcare Hack** and **Google–Build with AI** at HackCanada 2026.
- Architected a **FastAPI** rules-based engine that evaluates patient data (age, insurance, lab values) against a decision graph to auto-trigger voice calls, SMS, and referrals with zero clinician review.
- Engineered an **ElevenLabs + Twilio** outbound call pipeline with webhook-driven status sync.
- Auto-booked **Google Calendar** appointments upon patient confirmation in real time.
- Built a **React Flow** drag-and-drop builder for no-code clinical workflow design.

Software Engineer Intern

Jul. 2026 – Sep. 2026

Islamic Center of Markham – Masjid Darul Iman

Toronto, ON

- Engineered a **tilt-compensated digital compass** by fusing **accelerometer**, **gyroscope**, and **magnetometer** data via vector-based orientation math to deliver reliable real-time heading estimates at **60Hz**.
- Debugged and resolved a **platform-specific accelerometer sign-convention bug** and inconsistent magnetic declination timeouts on Android, restoring reliable heading accuracy across devices.
- Added **confidence-gated dip-angle filtering** to fix heading accuracy at high-latitude locations, then implemented **AsyncStorage** caching so the app stays reliable without a live network connection.
- Built a live announcements feed on the **WordPress REST API**, sanitizing HTML and stripping CMS markup so third-party content renders reliably, with pull-to-refresh support.
- Built a content-reader with structured multi-section indexing (**114 entries**) and persistent read-position tracking, then migrated push notifications to native **Expo Notifications** to cut vendor dependency.
- Unified visual design across native and **WebView**-rendered components via injected custom CSS.

Software Systems Intern

Jul. 2025 – Aug. 2025

Islamic Center of Markham – Masjid Darul Iman

Toronto, ON

- Developed a community-facing mobile application using **React Native (Expo)**, serving **1,000+ members** with real-time announcements and prayer schedules.
- Implemented push notifications with **OneSignal** and live data syncing via **Supabase**, enhancing user engagement.
- Built a **full-stack Volunteer Scheduling platform** with **React.js** and **Supabase**, coordinating **250+ volunteers**.
- Integrated secure **Supabase Auth** with **PostgreSQL**, enabling role-based access and seamless **CRUD** operations.

HACKATHONS & PROJECTS

☑ *Hampy (Hack the North 2026) | YOLOv8, BB-SLAM, BracketBot, FastAPI*

Sep. 2026

- Mapped **Meta Quest 3** hand tracking to the arm to teleoperate pick-and-place and record demonstrations.
- Trained a lightweight **YOLOv8** detector to locate and target the requested item before the pick.
- Implemented **BB-SLAM mapping** for end-to-end autonomous navigation between pickup and drop-off zones.
- Wired the full pipeline — **SLAM nav** → **YOLOv8** → learned pick → drive → release — behind a single API call.

☑ *Mimicoo (Hack the Valley 2025 - 6th Place) | Python, React, Librosa, Gemini API*

Oct. 2025

- Built an **AI pronunciation trainer** leveraging **Parselmouth** for infant speech disorder risk analysis.
- Designed a **REST API** architecture connecting audio-processing backend to a responsive, low-latency frontend dashboard.
- Deployed a full-stack **React/TypeScript/FastAPI** app delivering real-time fluency scoring and feedback.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript/TypeScript, SQL, HTML, CSS, R

Frameworks: React, React Native (Expo), Next.js, FastAPI, Spring Boot, Tailwind CSS, MediaPipe, OpenCV, PyTorch

Databases: PostgreSQL, Supabase, SQL (joins, indexing, CTEs, window functions)

Developer Tools: Git, GitHub Actions, Docker, Firebase, Google Cloud Platform, Vercel, Postman, VS Code, PyCharm